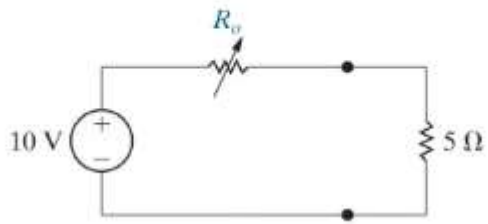
Problems II - Week 7

- Source Transformations
- Thevenin and Norton Equivalents

4.84 **a.** Find the value of the variable resistor R_o in the circuit in Fig. P4.84 that will result in maximum power dissipation in the 5 Ω resistor. (*Hint:* Hasty conclusions could be hazardous to your career.)

b. What is the maximum power that can be delivered to the 5 Ω resistor?

Figure P4.84

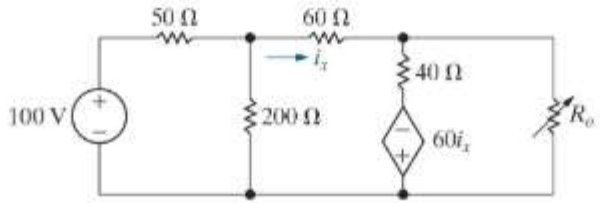


[a] Since $0 \leq R_o \leq \infty$ maximum power will be delivered to the 5 Ω resistor when $R_o = 0$.

[b] $P = \dfrac{10^2}{5} = 20$ W.

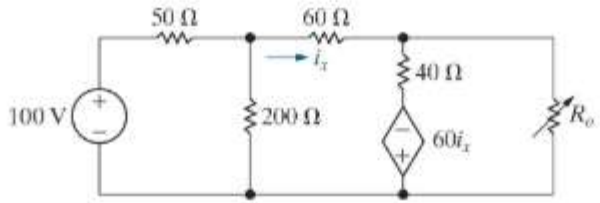
4.88 The variable resistor R_o in the circuit in Fig. P4.88 is adjusted until the power dissipated in the resistor is 1.5 W. Find the values of R_o that satisfy this condition.

Figure P4.88

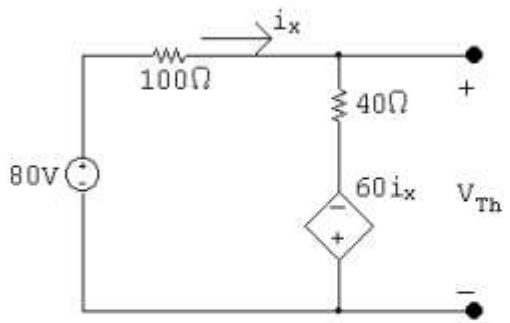


4.88 The variable resistor R_o in the circuit in Fig. P4.88 is adjusted until the power dissipated in the resistor is 1.5 W. Find the values of R_o that satisfy this condition.

Figure P4.88



We begin by finding the Thévenin equivalent with respect to R_o. After making a couple of source transformations the circuit simplifies to

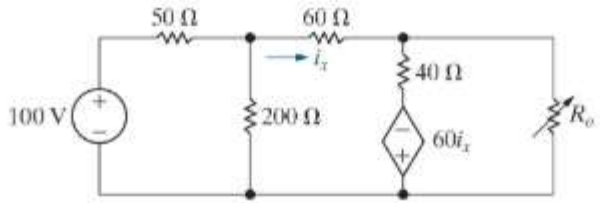


$$i_x = \frac{80 + 60i_x}{140}; \qquad i_x = 1\,\text{A};$$

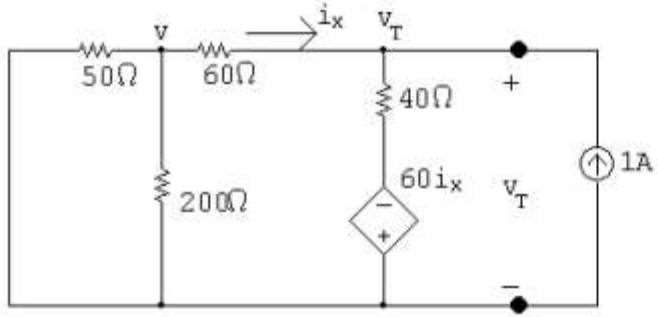
$$V_{\text{Th}} = 40i_x - 60i_x = -20i_x = -20\,\text{V}.$$

4.88 The variable resistor R_o in the circuit in Fig. P4.88 is adjusted until the power dissipated in the resistor is 1.5 W. Find the values of R_o that satisfy this condition.

Figure P4.88



Using the test-source method to find the Thévenin resistance gives



Use the node voltage method:

$$\frac{v}{50}+\frac{v-v_T}{60}+\frac{v}{200}=0;$$

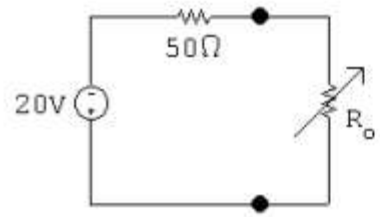
$$\frac{v_T-v}{60}+\frac{v_T+60i_x}{40}-1=0;$$

$$i_x=\frac{v-v_T}{60};$$

Solving, $v_T = 50$ V.

$$R_{Th}=\frac{v_T}{1\,\text{A}}=50\,\Omega.$$

Thus our problem is reduced to analyzing the circuit shown below.



$$\left(\frac{-20}{50+R_o}\right)^2 R_o=1.5;$$

$$\frac{400R_o}{R_o^2+100R_o+2500}=1.5;$$

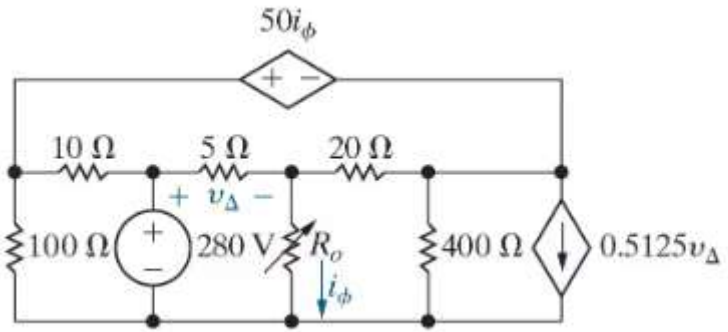
$$1.5R_o^2-250R_o+3750=0;$$

$$\therefore\quad R_o=16.67\,\Omega;\quad R_o=150\,\Omega.$$

4.90 The variable resistor in the circuit in Fig. P4.90 is adjusted for maximum power transfer to R_o.

a. Find the numerical value of R_o.

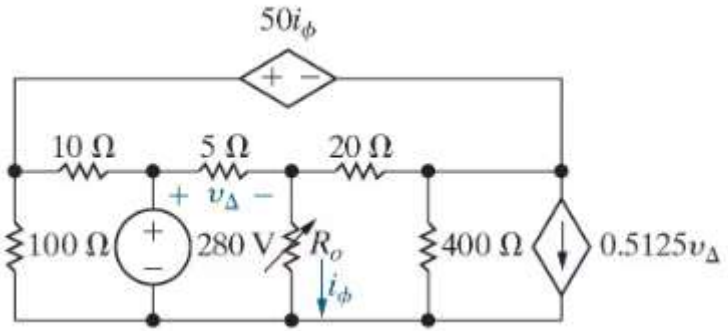
b. Find the maximum power delivered to R_o.



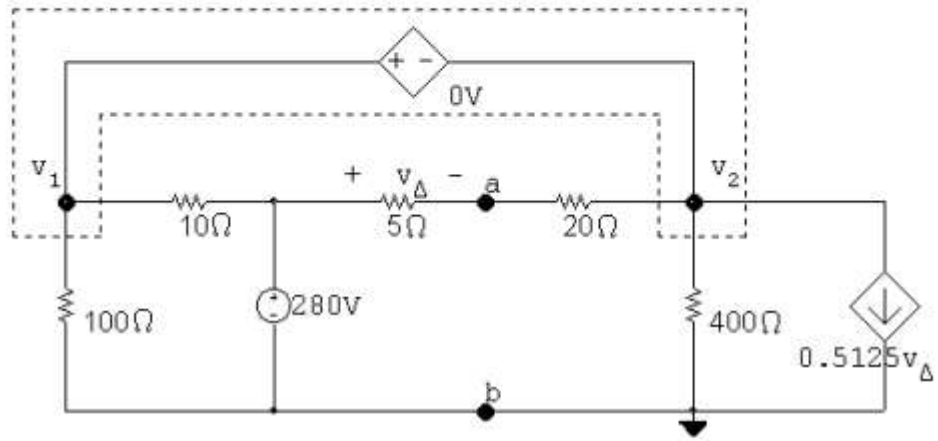
4.90 The variable resistor in the circuit in Fig. P4.90 is adjusted for maximum power transfer to R_o.

a. Find the numerical value of R_o.

b. Find the maximum power delivered to R_o.



First find the Thévenin equivalent with respect to R_o.
Open circuit voltage: $i_\phi = 0$; $50i_\phi = 0$.



$$\frac{v_1}{100} + \frac{v_1 - 280}{10} + \frac{v_1 - 280}{25} + \frac{v_1}{400} + 0.5125v_\Delta = 0;$$

$$v_\Delta = \frac{(280 - v_1)}{25}5 = 56 - 0.2v_1;$$

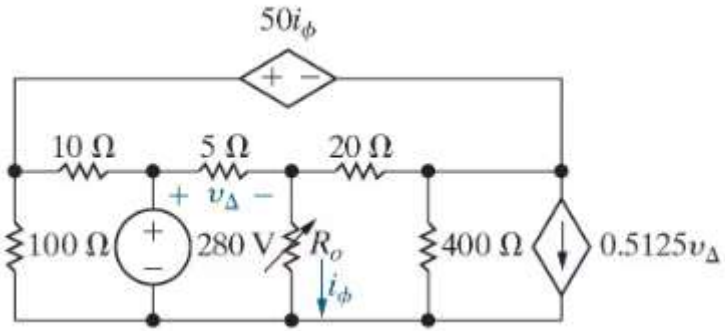
$$v_1 = 210\,\text{V}; \qquad v_\Delta = 14\,\text{V};$$

$$V_{\text{Th}} = 280 - v_\Delta = 280 - 14 = 266\,\text{V}.$$

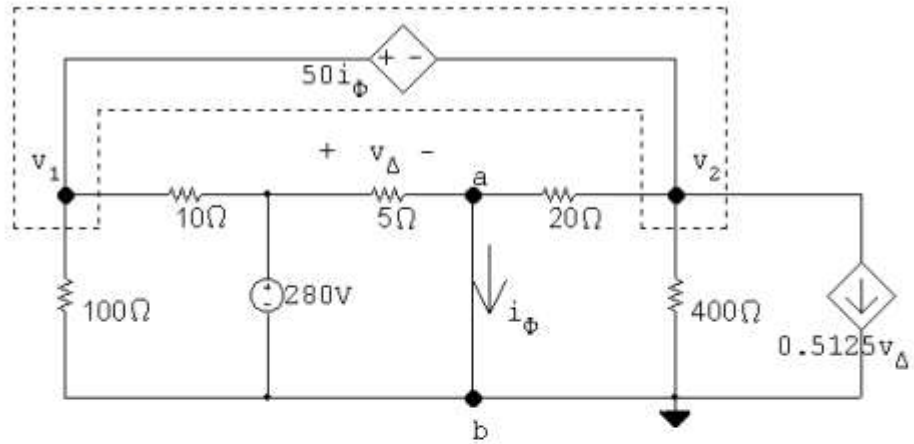
4.90 The variable resistor in the circuit in Fig. P4.90 is adjusted for maximum power transfer to R_o.

a. Find the numerical value of R_o.

b. Find the maximum power delivered to R_o.



Short circuit current



$$\frac{v_1}{100}+\frac{v_1-280}{10}+\frac{v_2}{20}+\frac{v_2}{400}+0.5125(280)=0;$$

$$v_\Delta = 280\,\text{V};$$

$$v_2 + 50i_\phi = v_1;$$

$$i_\phi = \frac{280}{5}+\frac{v_2}{20} = 56 + 0.05v_2;$$

$$v_2 = -968\text{V}; \qquad v_1 = -588\,\text{V};$$

$$i_\phi = i_{sc} = 56 + 0.05(-968) = 7.6\,\text{A};$$

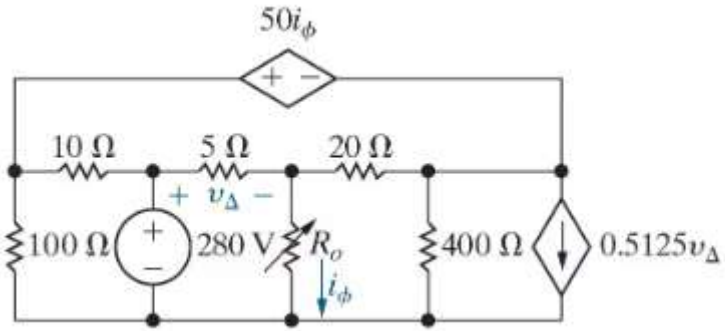
$$R_{Th} = V_{Th}/i_{sc} = 266/7.6 = 35\,\Omega;$$

$$\therefore \quad R_o = 35\,\Omega.$$

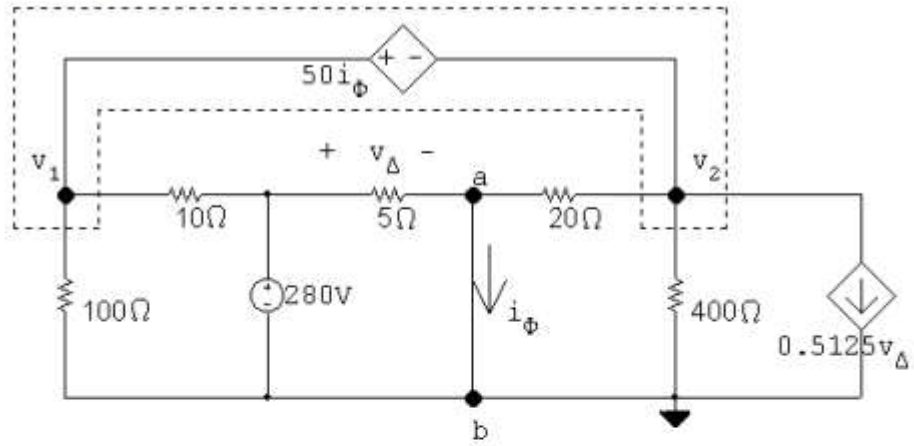
4.90 The variable resistor in the circuit in Fig. P4.90 is adjusted for maximum power transfer to R_o.

a. Find the numerical value of R_o.

b. Find the maximum power delivered to R_o.



Short circuit current



$$\frac{v_1}{100}+\frac{v_1-280}{10}+\frac{v_2}{20}+\frac{v_2}{400}+0.5125(280)=0;$$

$$v_\Delta = 280\,\text{V};$$

$$v_2 + 50i_\phi = v_1;$$

$$i_\phi = \frac{280}{5}+\frac{v_2}{20} = 56+0.05v_2;$$

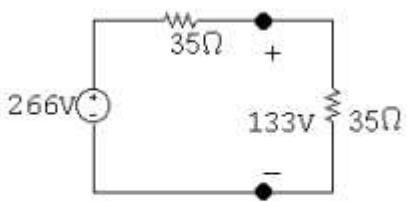
$$v_2 = -968\text{V}; \qquad v_1 = -588\,\text{V};$$

$$i_\phi = i_{sc} = 56+0.05(-968) = 7.6\,\text{A};$$

$$R_{Th} = V_{Th}/i_{sc} = 266/7.6 = 35\,\Omega;$$

$$\therefore \quad R_o = 35\,\Omega.$$

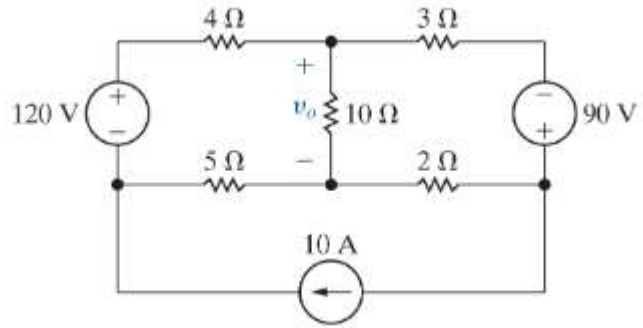
[b]



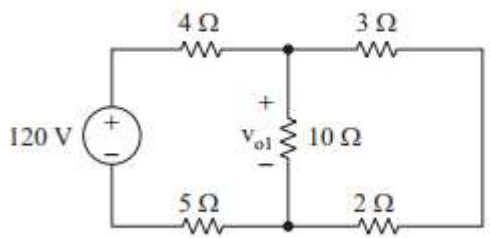
$$p_{\max} = (133)^2/35 = 505.4\,\text{W}.$$

4.93 Using the principle of superposition, find the current through the 10 Ω resistor in the circuit in Fig. P4.93.

Figure P4.93



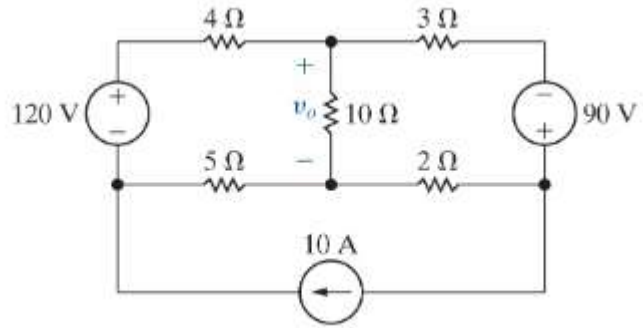
120 V source acting alone:



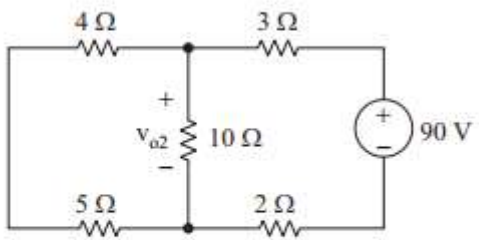
$$v_{o1} = \frac{10||14}{(4+5)+(10||4)} = \frac{2.8571}{11.8571}(120) = 28.9 \text{ V}$$

4.93 Using the principle of superposition, find the current through the 10 Ω resistor in the circuit in Fig. P4.93.

Figure P4.93



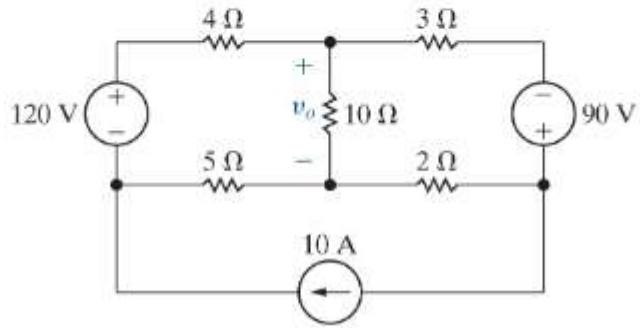
90 V source acting alone:



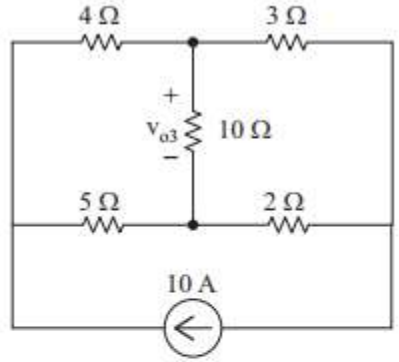
$$v_{o2} = \frac{10||9}{2+3+(10||9)}(-90) = -43.776\text{V}$$

4.93 Using the principle of superposition, find the current through the 10 Ω resistor in the circuit in Fig. P4.93.

Figure P4.93



10 A current source acting alone:



$$\frac{v_1 - v_2}{4} + \frac{v_1}{5} - 10 = 0$$

$$\frac{v_2 - v_1}{4} + \frac{v_2}{10} + \frac{v_2 - v_3}{3} = 0$$

$$\frac{v_3 - v_2}{3} + \frac{v_3}{2} + 10 = 0$$

Solving,

$$0.45v_1 - 0.25v_2 = 10$$

$$-0.25v_1 + 0.6833v_2 - 0.33v_3 = 0$$

$$0 - 0.33v_2 + 0.833v_3 = -10$$

$$v_1 = -5.3202 \text{ V}$$

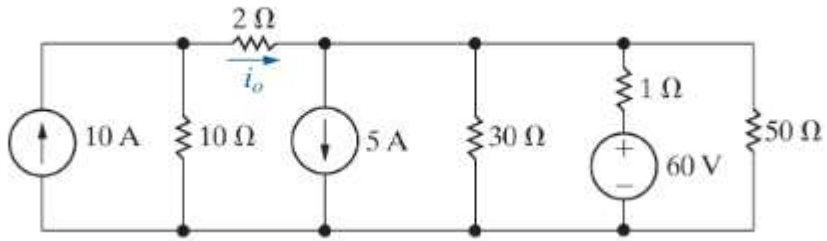
$$v_2 = -9.576 \text{ V} = v_{03}$$

$$v_3 = -15.796 \text{ V}$$

$$v_o = v_{o1} + v_{o2} + v_{o3} = -24.452 \text{ V}$$

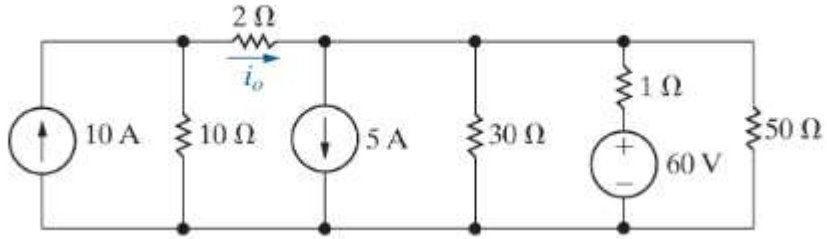
4.94 Using the principle of superposition, find the current through the 50 Ω resistor in the circuit shown in Fig. P4.94.

Figure P4.94

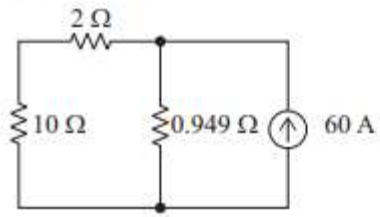


4.94 Using the principle of superposition, find the current through the 50 Ω resistor in the circuit shown in Fig. P4.94.

Figure P4.94



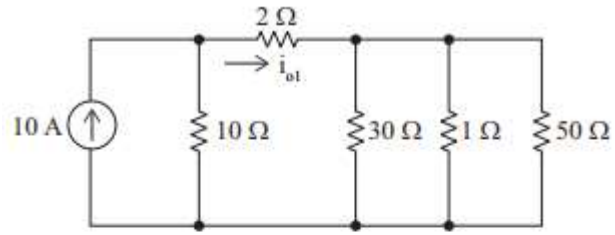
60 V source:



$$i_{o3} = \frac{-0.949}{11}(60) = -5.176 \text{ A}$$

$$i_o = i_{o1} + i_{o2} + i_{o3} = 4.345 \text{ A}$$

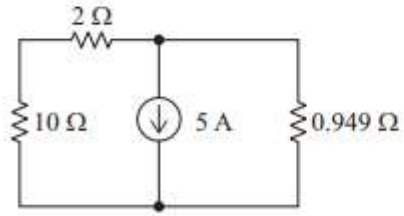
10 A source:



$$50||1||30 = 0.949 \ \Omega$$

$$\therefore \ i_{o1} = \frac{10}{10 + (1)}(10) = 9.09 \text{ A}$$

5 A source:



$$i_{o2} = \frac{0.949}{11}(5) = 0.4315 \text{ A}$$